

Recursion Practice Problems

1. Write a recursive function to calculate the factorial of a given number.
 - Sample Input: 5
 - Sample Output: 120
2. Write a recursive function to find the nth Fibonacci number.
 - Sample Input: 6
 - Sample Output: 8 (Fibonacci series: 0, 1, 1, 2, 3, 5, 8, 13,...)
3. Write a recursive function to check whether a given string is a palindrome or not.
 - Sample Input: "madam"
 - Sample Output: Palindrome
4. Write a recursive function to print all numbers from 1 to n.
 - Sample Input: 5
 - Sample Output: 1 2 3 4 5
5. Write a recursive function to calculate the sum of all digits of a given number.
 - Sample Input: 1234
 - Sample Output: 10 ($1 + 2 + 3 + 4 = 10$)
6. Write a recursive function to find the greatest common divisor (GCD) of two numbers.
 - Sample Input: 36, 60
 - Sample Output: 12
7. Write a recursive function to print the reverse of a given string.
 - Sample Input: "hello"
 - Sample Output: "olleh"
8. Write a recursive function to calculate the power of a number (x raised to the power y).
 - Sample Input: 2, 3
 - Sample Output: 8 ($2^3 = 8$)
9. Write a recursive function to find the sum of an array of integers.
 - Sample Input: {1, 2, 3, 4, 5}
 - Sample Output: 15
10. Write a recursive function to count the number of digits in a positive integer.
 - Sample Input:
12345
 - Sample Output:
5